

Comparing Auditable Architectures for LLM Agents

This term paper will compare three approaches to building auditable LLM agents: log-primary architectures, checkpoint-based architectures, and provenance-overlay systems. The study will draw on multiple recent works representing each of these approaches, without focusing on any single system as the primary reference, and will instead treat representative architectures from each category as equally important for comparison.

The paper will not assume that one architecture is universally more auditable. Instead, it will study how different architectures support different audit goals, such as identifying the source of an error, tracing claims back to evidence, reconstructing an execution, checking policy compliance, and reviewing agent decisions.

The main research question is:

How do log-primary, checkpoint-based, and provenance-overlay architectures differ in their support for specific audit tasks, and what trade-offs do they introduce in terms of storage, implementation complexity, privacy, and computational cost?

The evaluation will be based on a controlled agent workflow applied consistently across the different architectures, allowing for a structured comparison of how each system supports key audit tasks such as failure localization, evidence tracing, execution reconstruction, and counterfactual analysis.

The expected outcome is not a single ranking of architectures, but a clearer understanding of which architecture is most suitable for which audit requirement and use case.